

	Marks	Credits
L T P	Exam : 50	2
- - 2	Sessional : 50	
	Total :100	

Course Outcomes:

Upon completion of this course, students will be able to do the following:

- CO1 Analyze the asymptotic performance of algorithms.
- CO2 Write rigorous correctness proofs for algorithms.
- CO3 Demonstrate a familiarity with major algorithms and data structures.
- CO4 Apply important algorithmic design paradigms and methods of analysis.
- CO5 Synthesize efficient algorithms in common engineering design situations

A student has to perform 10-12 practical's based on theory paper.

Suggested list of experiments:

1. Write a program to search an element in a two -dimensional array using linear search.
2. Using iteration & recursion concepts write programs for finding the element in the array using Binary Search Method
3. Write a program to perform following operations on tables using functions only
4. Addition b) Subtraction c) Multiplication d) Transpose
5. Using iteration & recursion concepts write the programs for Quick Sort Technique
6. Write a program to implement the various operations on string such as length of string concatenation, reverse of a string & copy of a string to another.
7. Write a program for swapping of two numbers using call by value and call by reference strategies.
8. Write a program to implement binary search tree. (Insertion and Deletion in Binary search Tree)
9. Write a program to create a linked list & perform operations such as insert, delete, update, reverse in the link list
10. Write the program for implementation of a file and performing operations such as insert, delete, update a record in the file.
11. Create a linked list and perform the following operations on it
 - a) Add anode
 - b) Delete a node
12. Write a program to simulate the various searching & sorting algorithms and compare their timings for a list of 1000 elements.
13. Write a program to simulate the various graph traversing algorithms.
14. Write a program which simulates the various tree traversal algorithms.